

RegTrace-Agent: A Context-to-Feedback Framework for Evidence-Grounded Regulatory Response Review

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Abstract

Financial LLM systems are increasingly asked to judge whether an institution’s response actually resolves a prior review request, yet most financial benchmarks still evaluate static QA, extraction, or classification. We introduce RegTrace-Agent, a context-to-feedback framework for evidence-grounded response resolution. Given a reviewer request, an organizational response, and revised artifacts, the framework retrieves evidence, decomposes obligations, verifies visible gaps, converts adjudicated failures into scalar/category/natural-language feedback, and trains a guarded reviewer policy. We instantiate the framework on SEC comment-letter review and construct RegTrace-SEC, 472 public EDGAR examples with amended-filing evidence, visible-evidence labels, feedback fields, and later same-topic SEC follow-up signals. Experiments show that amended evidence is necessary, full natural-language feedback is the strongest reviewer-training signal among handwritten prompting, MIPROv2, and scalar/category GEPA variants, and a guarded verifier derived from the learned policy improves further. Later SEC follow-up comments provide external validation for the benchmark labels, suggesting that regulator-style feedback is a useful learning signal for evidence-grounded compliance reviewers.

1 Introduction

Financial NLP has rapidly moved toward LLM systems for long documents, domain-specific reasoning, and decision support. Models and benchmarks such as BloombergGPT, FinGPT, FinBen, FinanceBench, FinQA, TAT-QA, and ConvFinQA show both the promise of financial LLMs and the continuing difficulty of grounding answers in financial evidence (Wu et al., 2023; Yang et al., 2023; Xie et al., 2024; Islam et al., 2023; Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022). Recent FinNLP work has similarly emphasized retrieval-augmented financial QA, SEC filing QA, financial

intelligence benchmarks, factuality judging, and evasive-answer detection (Lithgow-Serrano et al., 2025; Lai et al., 2025; Klimaszewski et al., 2025; Wu et al., 2025; Al Nuaimi et al., 2025). Most such settings, however, still present a static input and ask for a static answer, label, or report.

Many operational finance and compliance workflows have a different structure. A regulator, auditor, lender, or control function requests a correction, explanation, disclosure, or supporting artifact; the organization responds that it has revised an external document; the reviewer must decide whether visible evidence actually satisfies the request. For example, a company may reply that it revised a filing to address a requested forecast disclosure, while the amended filing only shows combined revenue projections and still omits the requested entity-level assumptions. The task is therefore not to judge whether the response sounds cooperative, but whether the repaired artifact visibly closes the original gap. We call this class of problems evidence-grounded response resolution. It is related to fact verification and document-level legal NLI (Thorne et al., 2018; Koreeda and Manning, 2021), but adds a prior reviewer request, a response that may only claim a revision, revised evidence that must be checked, and textual feedback about what remains missing.

Existing work leaves three limitations for this setting. First, benchmarks rarely ask whether an organization has repaired a documented deficiency. Second, response evaluation often has weak obligation grounding: a response can sound persuasive while amended evidence omits a named disclosure, quantification, exhibit, legal basis, or accounting analysis. Third, prompt and pipeline optimization often discards feedback structure by receiving only scalar success, although review failures are naturally described in text: what was requested, what was claimed, what evidence showed, and what gap remained.

We study these limitations through U.S. Securities and Exchange Commission (SEC) comment-letter review. SEC staff issue comments requesting expanded disclosure, clarification, revised accounting discussion, removal of unsupported statements, or additional exhibits; companies respond in writing, often with statements such as “the disclosure has been revised accordingly.” After review completion, the SEC releases staff comment letters and company responses through EDGAR. These traces often point to amended filings and sometimes include later same-topic SEC follow-up comments. Accounting research treats comment letters as economically meaningful regulatory signals (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmardebeck, 2023; Ryans, 2021); for NLP, they provide observable request-response-evidence-feedback traces.

We introduce RegTrace-Agent, a context-to-feedback framework for evidence-grounded reviewers. It reconstructs review threads, retrieves revised-document evidence, decomposes requests into obligations, adjudicates visible evidence gaps, compiles scalar/category/full feedback, and optimizes a guarded reviewer policy. The learned reviewer is not a single hand-tuned prompt; it is the policy inside a system that gathers evidence, checks obligations, emits a guarded verdict, and uses adjudicated failures to improve the next offline reviewer.

This design makes response resolution a strong testbed for language-feedback optimization. Prompt-programming work treats LLM pipelines as declarative programs whose prompts and demonstrations can be optimized (Khatab et al., 2023); GEPA argues that natural-language feedback can expose reusable failure structure that scalar reward discards (Agrawal et al., 2026). In our setting, that structure is regulator-style: a mistake can be explained as a missing requested disclosure, unsupported revision claim, absent exhibit, or incomplete analysis.

The above limitations motivate three research questions. RQ1 asks whether public review traces can be converted into a benchmark for evidence-grounded response resolution. RQ2 asks whether full natural-language feedback is a stronger reviewer-training signal than scalar prompt search and coarse categories. RQ3 asks whether the resulting decisions generalize across threads, time, and issue categories, and whether unresolved decisions are corroborated by later regulator follow-up.

We make four contributions. First, we define evidence-grounded response resolution and introduce RegTrace-Agent, a context-to-feedback framework for evidence retrieval, obligation checking, feedback construction, and reviewer optimization. Second, we instantiate it as RegTrace-SEC, a public EDGAR benchmark with SEC comments, company responses, amended-filing evidence, labels, and feedback fields. Third, we show that full natural-language feedback is the strongest reviewer-training signal, outperforming handwritten prompting, MIPROv2, scalar/category GEPA variants, and a supervised lexical baseline; a guarded verifier distilled from the learned policy improves further. Fourth, we provide evidence ablations, independent-feedback and feedback-component ablations, prompt-evolution analysis, and systematic corroboration from real next-round SEC follow-up comments.

2 Related Work

Financial NLP and evidence-grounded evaluation. Financial NLP has moved from broad document classification toward long-document reasoning, evidence grounding, and compliance-sensitive decisions. FinQA, TAT-QA, and ConvFinQA introduced financial reasoning questions over reports, tables, and conversations (Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022); FinanceBench evaluates open-book QA over company documents (Islam et al., 2023). Recent FinNLP work has similarly emphasized RAG over financial documents, accessible finance-intelligence benchmarks, factuality and causality judging, SEC filing QA, and evasive-answer detection (Lithgow-Serrano et al., 2025; Klimaszewski et al., 2025; Lai et al., 2025; Al Nuaimi et al., 2025; Wu et al., 2025). Our task differs by asking whether retrieved evidence satisfies a prior regulator request.

SEC comment letters and regulatory review. Accounting and disclosure research has studied SEC comment letters as regulatory signals, including remediation costs, disclosure effects, recurring disclosure deficiencies, and textual classification of comment-letter topics (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmardebeck, 2023; Ryans, 2021). That literature generally treats comment letters as economic or disclosure events. We instead treat the correspondence as document-grounded supervision for LLM systems. Each review unit contains a staff request, company

187 response, amended filing evidence, and sometimes
188 a next-round SEC follow-up.

189 **Prompt optimization and language feedback.**
190 DSPy reframes LLM pipelines as declarative pro-
191 grams whose prompts and demonstrations can be
192 optimized against metrics (Khatab et al., 2023).
193 MIPROv2 and related optimizers primarily use
194 scalar feedback. A broader line studies language
195 feedback, including self-refinement, verbal rein-
196 forcement, and critique-based training (Scheurer
197 et al., 2022; Madaan et al., 2023; Shinn et al., 2023).
198 GEPA uses natural-language reflection over execu-
199 tion traces (Agrawal et al., 2026). This distinction
200 is central here because SEC response errors are
201 naturally textual: missing disclosures, unsupported
202 revision claims, absent exhibits, incomplete analy-
203 sis, or over-demand beyond the SEC request.

204 3 RegTrace-Agent Framework

205 3.1 General Review-Trace Setting

206 RegTrace-Agent applies to workflows where a re-
207 sponse claims to satisfy a prior review request by
208 modifying or supplying evidence outside the re-
209 sponse itself. Formally, each review trace contains
210 a request q , a response r , a collection of revised
211 or supporting artifacts $\mathcal{D}$, and optional later re-
212 viewer feedback u^+ . The system retrieves evidence
213 $e = R(q, r, \mathcal{D})$, extracts obligations $o_{1:m} = O(q)$,
214 and applies a verifier $V(q, r, e, o_{1:m})$ whose output
215 is either resolved or unresolved. The same trace
216 can also produce feedback for learning. A scalar
217 signal says whether the prediction is correct. A
218 category signal names a broad issue or gap type. A
219 full natural-language signal describes the request,
220 the response action, the visible evidence, and any
221 unmet requirement. This makes the framework
222 applicable beyond SEC review whenever request-
223 response-evidence traces are available.

224 3.2 SEC Instantiation

225 Each SEC example contains three test-time
226 fields: `sec_comment`, `company_response`, and
227 `retrieved_snippets` from amended filings. The
228 output is binary. `resolved` means visible amended
229 evidence covers the material request; `unresolved`
230 means a named item, quantification, exhibit, docu-
231 ment, legal analysis, accounting analysis, or other
232 material requirement remains missing. The label is
233 a visible-evidence judgment, not a legal determina-
234 tion or an inference from absence of later follow-
235 up.

Quantity	Count
Examples	472
SEC review threads	109
Companies	95
Resolved / unresolved	153 / 319
Examples with verified follow-up	236
Issue categories	7
Visible-evidence gap types	8

Table 1: Benchmark scale. Issue categories include other disclosure issues, risk factors, SPAC disclosure, revenue recognition, crypto, non-GAAP, and cybersecurity.

507 3.3 SEC Trace Construction

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Figure 1 separates the runtime reviewer from the
feedback-learning loop. In the SEC instantiation,
we harvest UPLOAD and CORRESP materials from
EDGAR, group staff comments and company re-
sponses into review threads, and select cases where
the company claims a disclosure was added, re-
vised, removed, or filed elsewhere. We identify
amended filing candidates using filing metadata,
amendment references, page and section cues, and
response text. Candidate filings are drawn from
a window of 10 days before to 60 days after the
response date, with at most 8 candidate filings per
row. A local lexical ranker retrieves top chunks,
and an LLM reranker/adjudicator evaluates the top
3 snippets. From 589 revision-claim cases, 588
have candidate filings and snippets; 472 remain
after requiring visible evidence that directly or par-
tially addresses the SEC request. We then adjudi-
cate whether visible evidence satisfies the request
and construct labels plus feedback fields.

The retrieval component gives the agent evi-
dence to inspect. The adjudication and feedback
layer gives it something to learn from: what the
reviewer asked for, what the company claimed to
do, what the visible evidence supports, and what
remains missing. These adjudication fields are not
shown as ordinary test-time inputs. They are used
for optimization feedback, audits, and analysis.

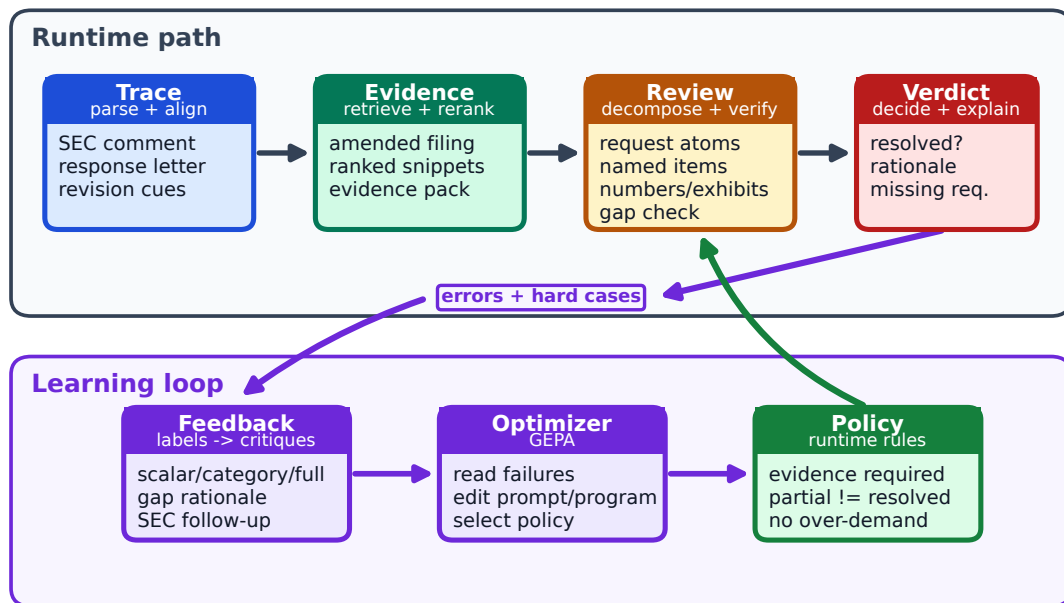
507 3.4 Scale, Categories, and Splits

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RegTrace-SEC contains 472 examples from 109
SEC review threads and 95 companies. The bench-
mark focuses on difficult revision-claim cases, and
the label distribution is operationally realistic: 153
examples are resolved, and 319 are unresolved.
Table 1 summarizes the corpus and split regimes.

The corpus covers seven issue categories and

RegTrace-Agent: closed-loop evidence review and feedback optimization



Test-time inputs: SEC comment, company response, retrieved amended evidence. Feedback fields are train/audit only.

Figure 1: RegTrace-Agent architecture. The runtime path parses a review trace, retrieves amended-filing evidence, checks request obligations, and emits a guarded verdict. The learning loop stores scalar, category, and natural-language feedback, lets GEPA revise the reviewer policy, and reinjects the learned policy into the obligation-review step. At test time the reviewer sees only the SEC comment, company response, and retrieved amended evidence.

273 eight visible-evidence gap types; Appendix B gives
 274 the full distribution. We use grouped, temporal, and
 275 topic-holdout regimes. The grouped split holds out
 276 entire review threads (229/104/139 train/dev/test).
 277 The temporal split trains on 2022–2023 and tests
 278 on 2024–2025 (264/64/144). The topic-holdout
 279 split tests transfer to held-out issue categories
 280 (279/68/125), including crypto and non-GAAP
 281 cases absent from training categories.

282 3.5 From Labels to Feedback

283 A standard benchmark would stop at binary labels.
 284 Our construction preserves the textual structure of
 285 each decision. For every example, adjudication
 286 records the material SEC request, the company
 287 action, evidence relevance, supporting evidence or
 288 resolution basis, visible unmet requirement, and a
 289 short rationale. This lets us compare increasingly
 290 rich optimization signals:

- 291 • scalar: correct or incorrect;
- 292 • category: correct/incorrect plus coarse issue or
 293 gap type;
- 294 • full: natural-language feedback describing the
 295 SEC request, company action, evidence coverage,

and unmet requirement.

Full feedback can state that the evidence discusses
 combined revenue projections but fails to disclose
 entity-level operating costs, product pricing, gross
 margins, and forecast limitations. This is the
 reusable decision rule that scalar metrics discard.

4 Methods

4.1 Prediction Setup

The prediction model receives the SEC comment,
 company response, and raw retrieved amended-
 filing snippets. It returns a binary label and a short
 rationale. Explicit resolved is scored as resolved.
 unresolved, partially_resolved, and other in-
 sufficient labels are normalized to unresolved. The
 rationale is required for inspection but not directly
 rewarded.

4.2 Optimization Ladder

We compare reviewer-training signals rather than
 evidence access. Baseline prompt is a hand-
 written reviewer instruction. MIPROv2 is scalar
 prompt optimization. GEPA-scalar uses reflective
 prompt optimization with correctness-only feed-
 back. GEPA-category adds coarse category feed-

back. GEPA-full uses full natural-language adjudication feedback. We also include TF-IDF, a frozen sentence-encoder scoring head, a Jev typed decision gate, and an independent-feedback GEPA ablation for reviewer-facing robustness.

All methods receive the same evidence at test time, isolating the optimization signal rather than evidence access. The key comparison is whether natural-language feedback adds value beyond prompt search itself.

4.3 Implementation Details

All main prompt-optimization runs use openai/gpt-4o-mini as the frozen prediction model with temperature 0 and 500 output tokens. GEPA uses the same model as the reflection model with temperature 0.7 and 1500 reflection tokens, seed 17, max_metric_calls=150, reflection minibatch size 4, Pareto candidate selection, and 4 evaluation threads. MIPROv2 uses the same prediction and prompt models, 8 trials, 4 prompt candidates, and at most 4 labeled and 4 bootstrapped demonstrations. The frozen encoder scorer uses all-MiniLM-L6-v2 with a logistic head over request, response, evidence, and request-evidence matching features. The Jev gate uses a choice question over resolved/unresolved. The evidence reranker/adjudicator also uses gpt-4o-mini at temperature 0. Model aliases rather than dated provider snapshots are recorded in the released artifacts.

4.4 Optimization as Feedback-Conditioned Search

We view each method as search over a prompt or prompt program p . Given an input x and label y , the frozen LLM defines $\hat{y}_p(x) = f_\theta(x; p)$, and optimization seeks a prompt with high validation reward. The methods differ in what they observe after an error. Scalar optimization observes only correctness. Category feedback adds a coarse gap type. Full feedback adds a short critique describing the request, company action, visible evidence, and unmet requirement.

The intuition is information-theoretic. Let C be the latent cause of failure, such as missing quantification or unsupported revision claim. Full feedback should help when it lowers uncertainty about C , because the optimizer can propose a more targeted semantic edit. This is not a general guarantee: a noisy or misaligned language channel can hurt. The hypothesis of RegTrace-Agent is that review traces

Input	Acc.	Ma-F1	Res-F1	Unres-F1
Response only	0.712	0.575	0.333	0.817
+ raw snippets	0.784	0.748	0.651	0.844
+ oracle summary	0.964	0.956	0.938	0.975

Table 2: Evidence ablation. Adding raw amended-filing snippets raises macro-F1 by 17.3 points. The oracle summary is an upper bound, not a deployable baseline.

are unusually well suited to language feedback because the failure causes are naturally written as reviewer critiques.

4.5 What the Optimizer Sees

During optimization, each candidate prompt is evaluated on development examples. When a prediction is incorrect, the metric returns feedback in one of three forms:

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scalar: incorrect
category: incorrect;
gap_type=missing_specific_disclosure
full: incorrect. The SEC requested
X. The company claimed Y. The visible
evidence shows Z, but does not show
required item W.
```

Full feedback summarizes the adjudicated relationship between the SEC request, response, and retrieved evidence for optimization examples only. It does not reveal hidden evidence at test time; any advantage must come from a better learned checklist for visible-evidence review.

5 Experiments

5.1 Evidence Ablation

We first test whether amended-filing evidence matters. Table 2 evaluates a response-only condition, a raw-evidence condition, and an oracle evidence-summary condition on the grouped held-out test set.

Response-only prompting cannot verify whether a revision claim is supported. Raw snippets nearly double resolved F1, while the oracle summary shows that much of the remaining difficulty is interpreting snippets against a specific request.

5.2 Main Context-to-Feedback Results

Table 3 reports the main reviewer-training ladder on the frozen grouped split. All methods receive the same test-time context.

The majority baseline has high accuracy because unresolved examples dominate, but zero resolved F1. The handwritten prompt is conservative: it has perfect unresolved precision but misses most gaps.

Method	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Majority unresolved	0.691	0.409	0.000	0.817	0.691	1.000
Baseline prompt	0.424	0.402	0.518	0.286	1.000	0.167
MIPROv2	0.655	0.652	0.619	0.684	0.929	0.542
GEPA-scalar	0.705	0.675	0.577	0.773	0.824	0.729
GEPA-category	0.597	0.595	0.562	0.627	0.870	0.490
GEPA-full	0.784	0.756	0.674	0.839	0.867	0.812

Table 3: Frozen grouped-split leaderboard. GEPA-full gives the best macro-F1 and unresolved F1 under the same raw-evidence test-time context.

Check	Condition	Macro-F1
Lexical baseline	TF-IDF, response only	0.523
Lexical baseline	TF-IDF, raw snippets	0.523
Diagnostic upper bound	TF-IDF, oracle evidence summary	0.858
Scoring head	Frozen encoder, raw snippets	0.667
Decision gate	Jev typed choice, raw snippets	0.722
Feedback source	GEPA-full, original feedback	0.756
Feedback source	GEPA-full, independent feedback	0.750

Table 4: Robustness checks. The raw-snippet TF-IDF baseline shows that lexical supervision does not explain the main result. Frozen encoder and Jev decision scorers are cheap non-generative triage components. The independent-feedback ablation shows that GEPA-full does not depend on reusing the original adjudication rationales.

MIPROv2 improves substantially, GEPA-scalar improves recall, and GEPA-category is too coarse in this run. GEPA-full is strongest overall, improving macro-F1 by 10.4 points over MIPROv2 and 8.1 points over GEPA-scalar. This shows that full feedback is not merely adding more text; it trains the reviewer to compare requested obligations against visible evidence.

Table 4 addresses two alternate explanations and one deployment question. A TF-IDF logistic regression classifier trained on the same raw evidence reaches only 0.523 macro-F1, so lexical supervision does not explain the GEPA-full gain. A frozen sentence encoder with a trained logistic scoring head reaches 0.667 macro-F1 without generation, and a Jev typed decision gate reaches 0.722, showing that RegTrace-Agent can support cheap triage scorers, but still trails the full feedback-trained reviewer. To test possible feedback leakage, we regenerate full textual feedback with an independent gpt-5.4-mini writer that sees only test-time fields plus the gold label, not the original adjudication rationales; GEPA with this feedback reaches 0.750 macro-F1, nearly matching the original 0.756.

We also compute five-fold out-of-fold results over review-thread groups. Mean macro-F1 is

0.698 for GEPA-full, 0.647 for MIPROv2, 0.609 for GEPA-scalar, 0.545 for GEPA-category, and 0.446 for the handwritten baseline. The fold standard deviations are nontrivial (0.151 for GEPA-full and 0.121 for MIPROv2), so we treat the single grouped split as a headline comparison and the out-of-fold analysis as stability evidence rather than as a deterministic margin claim.

5.3 Why Full Feedback Helps

The advantage of GEPA-full is not simply risk aversion. Jev reaches very high unresolved recall (0.958) but lower resolved F1 (0.576), making it useful as a triage gate that flags likely unresolved responses. GEPA-full has lower unresolved recall but higher macro-F1, because it better balances skepticism with evidence sufficiency. It learns when a company revision claim is unsupported, but also when visible amended evidence already satisfies the SEC request.

The mechanism is visible in the feedback channel. Scalar feedback says only that a decision was wrong; category feedback names a coarse gap. Full feedback states what the SEC requested, what the company claimed, what the amended evidence showed, and what requirement remained unmet. This gives the optimizer a reusable review trace rather than a bare reward. The resulting prompt evolution shifts from generic caution to obligation-level comparison: identify material SEC request elements, inspect amended evidence for named disclosures, quantification, exhibits, and legal or accounting analysis, and distinguish partial disclosure from resolution.

The trade-off is cost and calibration. GEPA-full requires adjudicated natural-language feedback and offline optimization, so it is not the cheapest runtime gate. It can also become over-skeptical when evidence substantially satisfies a request but does not present the answer in the exact form the prompt expects. We therefore view Jev and frozen encoders as high-throughput first-pass gates, and

Reviewer	Macro-F1	Res-F1	Unres-F1
Monolithic reviewer	0.645	0.614	0.676
Obligation-guided reviewer	0.619	0.613	0.624
GEPA-full optimized prompt	0.756	0.674	0.839
RegTrace guarded verifier	0.801	0.727	0.874

Table 5: Reviewer-structure ablation. The guarded verifier is a policy-distillation prototype derived from the obligation-evidence policy exposed by GEPA-full prompt evolution.

GEPA-full as the deeper reviewer for difficult or high-impact cases.

5.4 Reviewer-Structure Ablation

The optimization ladder tests feedback richness. We also ask whether reviewer structure matters. Table 5 compares a monolithic judgment, an obligation-guided judgment, and a guarded verifier that requires every material obligation to be visibly satisfied, with GEPA-full as the optimized-prompt reference.

Simply listing obligations is not enough; the gain comes from the guard that requires visible support for each material request. We report this as a policy-distillation prototype rather than the primary optimizer comparison, since it adds human-designed structure after prompt-evolution analysis.

5.5 Feedback Component Ablation

Table 6 holds test-time input fixed and changes only optimization-time feedback.

Request-action-only feedback induces aggressive skepticism and the highest unresolved recall, but sacrifices resolved F1. Full feedback teaches both sides of the boundary: unmet requirements for unresolved cases and supporting evidence for resolved cases.

5.6 Temporal and Topic Generalization

Table 7 shows that the optimized prompt transfers beyond the grouped split. In the temporal setting, GEPA-full trains/tunes on 2022–2023 and tests on 2024–2025 material. In the topic setting, test examples include crypto and non-GAAP cases absent from training issue categories.

The topic-holdout result argues against a curated-topic explanation. Per-topic breakdowns show GEPA-full improves over MIPROv2 on crypto, non-GAAP, revenue recognition, risk factors, and broad other-disclosure cases, with largest gains in revenue recognition and non-GAAP.

6 External SEC Follow-Up Corroboration

Visible-evidence labels come from an adjudication pipeline, so we test whether they align with later regulator behavior. Verified same-topic next-round SEC follow-up is not the primary label, but it is a useful corroboration signal because it can restate or narrow an unresolved gap.

We additionally audit 60 examples with an independent LLM auditor that receives the SEC comment, company response, and visible amended evidence but not the original label rationale. The audit agrees with the benchmark label in 49 of 60 cases, with mean confidence 0.897. We report this as an audit of visible-evidence label consistency, not as expert legal/accounting validation.

Among 236 examples with verified follow-up, unresolved cases are corroborated at 68.1%, while resolved cases are corroborated at only 9.2% (Table 8). Follow-up availability itself is balanced across labels (49.7% for resolved and 50.2% for unresolved; Fisher exact $p = 1.0$), so this is not merely an availability artifact.

Following reviewer-style paired analysis, we also treat later follow-up as a noisy external reference on a clean 220-example subset. The benchmark label reaches 0.756 macro-F1 against this signal, compared with 0.590 for five-fold out-of-fold GEPA-full predictions and 0.460 for the handwritten baseline. We therefore use follow-up primarily as label-validity evidence, not as proof that every GEPA-full win is separately corroborated beyond the unresolved base rate.

In one expense-disclosure case, the SEC asked for material components and trend discussion. The company claimed a revision, but retrieved evidence showed only a line item and totals; GEPA-full predicted unresolved, and the next-round SEC follow-up again requested the component breakout. SPAC transaction-background cases show the same pattern for missing negotiation details.

7 Prompt Evolution and Case Studies

7.1 Prompt Evolution

Prompt evolution shows that GEPA-full does not merely make the prompt longer. It turns adjudication feedback into a compact regulatory reading policy. The seed prompt is a single sentence; the final GEPA-full run saves optimized candidates of 349 and 397 words (Appendix A).

Scalar and category feedback learn general caution: do not trust a revision claim unless evidence

Feedback	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Full feedback	0.784	0.756	0.674	0.839	0.867	0.812
No evidence rationale	0.712	0.682	0.583	0.780	0.826	0.740
No unmet requirement	0.741	0.708	0.609	0.806	0.833	0.781
Request + action only	0.784	0.724	0.595	0.853	0.806	0.906

Table 6: Feedback component ablation. Request-action feedback induces aggressive unresolved detection, while full feedback gives the best macro-F1 by calibrating skepticism with evidence rationales.

Split	Method	Acc.	Macro-F1	Unres-F1	Unres-R
Time	Baseline	0.451	0.434	0.336	0.206
	MIPROv2	0.521	0.519	0.489	0.340
	GEPA-scalar	0.653	0.650	0.679	0.546
	GEPA-full	0.826	0.791	0.877	0.918
Topic	Baseline	0.440	0.440	0.426	0.274
	MIPROv2	0.576	0.565	0.635	0.484
	GEPA-scalar	0.528	0.524	0.569	0.411
	GEPA-full	0.784	0.734	0.849	0.800

Table 7: Temporal and topic-holdout generalization. GEPA-full transfers substantially better than scalar optimization.

Group	N	D/P	Rate
Unres. + f/u	160	109	68.1%
Res. + f/u	76	7	9.2%

Table 8: Real follow-up corroboration. Later same-topic SEC comments directly or partially corroborate unresolved visible-evidence labels far more often than resolved labels.

Method	Claim insuff.	Evidence required	Request match	Missing items
GEPA-full	2/2	2/2	2/2	2/2
GEPA-scalar	2/2	2/2	0/2	0/2
GEPA-category	2/2	2/2	0/2	0/2
Req.+action only	2/2	2/2	2/2	2/2

Table 9: Rule presence in optimized prompt candidates. Full feedback more consistently induces named-request matching and missing-item reasoning.

supports it. Full feedback more consistently learns named-request matching and missing-item reasoning (Table 9), shifting from generic skepticism to obligation-level comparison: SEC request → company action → visible evidence → remaining gap.

7.2 Method-Disagreement Cases

The strongest GEPA-full wins involve concrete request-evidence mismatches. In a financial-projections case, the SEC requested individual-company forecast information and material assumptions; the amended evidence showed only combined revenue projections. GEPA-full marked it unresolved because the requested assumption set was still absent.

In a crypto terminology case, the amended evidence defined stablecoins but did not explain the SEC’s requested contrast with unstable cryptocurrency. Baseline, MIPROv2, GEPA-scalar, and GEPA-category predicted resolved; GEPA-full predicted unresolved. In a PIPE financing case, MIPROv2 treated the presence of price terms as sufficient; GEPA-full required equivalent price-per-share disclosure for each agreement, a gap later echoed in SEC follow-up.

GEPA-full also has informative errors. In one ownership-percentage case, retrieved evidence contained redemption-scenario calculations and the benchmark label was resolved. GEPA-full predicted unresolved because it wanted a more explicit cover-page presentation. This reveals the main calibration risk: full-feedback prompts learn useful skepticism but can over-require exact presentation. Appendix C lists additional cases.

8 Discussion

The main result is not simply that one optimizer beats another. RegTrace-Agent turns a recurring compliance pattern into an offline feedback-learning system: collect the trace, retrieve the revised artifact, check obligations against visible evidence, issue a guarded verdict, and recycle adjudicated failures as feedback. The learned reviewer can be used directly, distilled into a guarded verifier, or converted into lightweight scoring heads for high-throughput triage. The output is a traceable unresolved-gap rationale, not only a binary label.

611 Language feedback helps because it preserves
612 decision structure. A scalar signal says only that
613 a verdict failed; full feedback says which named
614 request, company action, amended snippet, and
615 remaining obligation were mismatched. The opti-
616 mized prompt therefore becomes an interpretable
617 policy: extract material requests, distrust unsup-
618 ported revision claims, inspect evidence for named
619 items, avoid over-requiring, and distinguish partial
620 disclosure from resolution.

621 SEC correspondence is the public setting where
622 we can observe request, response, revised filing,
623 and later reviewer behavior, but the trace structure
624 is broader. Audit remediation, regulatory report-
625 ing, lending-document stipulations, data-quality
626 exceptions, and internal compliance monitoring
627 also contain request-response-evidence loops. The
628 transferable claim is that such traces can support
629 retrieval, obligation checking, adjudication feed-
630 back, optimized reviewing, and hard-case mining;
631 the limitation is that high-recall language-feedback
632 policies can become over-skeptical.

633 **9 Conclusion**

634 We introduced RegTrace-Agent, a context-to-
635 feedback framework for evidence-grounded re-
636 sponse review, and instantiated it as RegTrace-SEC
637 using public SEC traces. Across grouped, tem-
638 poral, and topic-holdout evaluations, full natural-
639 language feedback yields the strongest reviewer.
640 Evidence ablations, prompt evolution, and SEC
641 follow-up corroboration suggest that financial NLP
642 benchmarks can move from static labels toward
643 feedback-rich review systems that preserve the rea-
644 sons behind expert decisions.

645 **Limitations and Data Statement**

646 This study is conditioned on visible retrieved ev-
647 idence, so missed snippets can affect labels. The
648 benchmark is LLM-adjudicated, with indepen-
649 dent audit and external follow-up corroboration
650 but without a full expert legal/accounting audit.
651 Same-topic SEC follow-up is a noisy behavioral
652 signal, not ground truth. We frame labels as
653 visible-evidence judgments rather than legal de-
654 terminations. An anonymized release will be pro-
655 vided at [https://anonymous.4open.science/](https://anonymous.4open.science/r/regtrace-sec-XXXX/)
656 [r/regtrace-sec-XXXX/](https://anonymous.4open.science/r/regtrace-sec-XXXX/); it includes benchmark
657 JSONL, splits, accession identifiers, retrieval meta-
658 data, snippets, labels, feedback fields, prompts, out-
659 puts, and analysis scripts.

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796	A Full Optimized Prompts	
797	The main paper quotes short excerpts from opti-	
798	mized prompts. Below are the full GEPA-full op-	
799	timized candidates exposed by the final grouped-	
800	split run. The seed prompt was: “Classify whether	
801	a company resolved an SEC comment using the	
802	visible evidence.”	
803	A.1 GEPA-full Optimized Candidate 1	
804	You are tasked with classifying whether a com-	
805	pany has resolved a comment from the SEC based	
806	on the provided evidence. Follow these instruc-	
807	tions carefully:	
808	1. Input Format: You will receive three types of	
809	inputs: <code>sec_comment</code> , <code>company_response</code> , and	
810	<code>amended_evidence</code> .	
811	2. Classification Criteria: Determine if the com-	
812	pany has fully resolved the SEC comment by ana-	
813	lyzing the company response in conjunction with	
814	the amended evidence. If the response satisfies	
815	the SEC’s request and the evidence supports this	
816	claim, classify the comment as resolved. If the re-	
817	sponse does not meet the SEC’s request or if there	
818	are still outstanding issues, classify the comment	
819	as unresolved.	
820	3. Evaluation Process: Carefully compare the	
821	SEC request against the company’s response and	
822	the amended evidence. Look for specific items or	
823	information requested by the SEC, quantification	
824	or detailed analysis that meets SEC requirements,	
825	references to regulatory guidance or previous de-	
826	cisions, and clear explanations of how the com-	
827	pany adjusted disclosures or practices. Take note	
828	of visible unmet requirements, partial fixes, or	
829	lack of supporting documentation in the amended	
830	evidence.	
	4. Output Format: Include a reasoning section,	831
	a label indicating resolved or unresolved, and a	832
	concise reason.	833
	5. Feedback Incorporation: Be vigilant about	834
	claims made by the company; ensure that	835
	amended evidence corroborates these claims.	836
	Identify instances where the response may par-	837
	tially address SEC concerns but still falls short of	838
	complete compliance.	839
	A.2 GEPA-full Optimized Candidate 2	840
	You are tasked with classifying whether a com-	841
	pany has resolved a comment issued by the SEC	842
	based on provided evidence. Follow these specific	843
	instructions for the task:	844
	1. Input Format: You will receive <code>sec_comment</code> ,	845
	<code>company_response</code> , and <code>amended_evidence</code> .	846
	The SEC comment is the original request for clar-	847
	ification or additional information. The company	848
	response indicates what changes were made or	849
	explanations provided. The amended evidence	850
	contains snippets from the amended filing or other	851
	documents that support the response.	852
	2. Classification Criteria: Determine whether the	853
	company’s response and amended evidence ade-	854
	quately address the SEC’s concerns as outlined	855
	in the original comment. A response is resolved	856
	if it fully satisfies the requirements of the SEC	857
	comment, including clear explanations, specific	858
	details, and quantifications when necessary. A	859
	response is unresolved if there are still gaps or	860
	missing elements in the company’s explanation or	861
	if amended evidence does not address all parts of	862
	the SEC comment.	863
	3. Evaluation Process: Carefully analyze the SEC	864
	comment and identify the specific details or expla-	865
	nations requested. Review the company response	866
	to see if the company acknowledged the comment	867
	and provided a meaningful response. Examine	868
	amended-evidence snippets to confirm that they	869
	support the company’s claims and cover all re-	870
	quested aspects of the SEC comment. Look for	871
	direct references in the amended evidence that	872
	address the SEC’s specific requests, such as own-	873
	ership structures, accounting adjustments, or en-	874
	forcement risks.	875
	4. Labeling: If the company has fully addressed	876
	the SEC’s concerns with adequate evidence, label	877
	the response resolved. If there are omissions or	878
	inadequacies in addressing SEC requests, label it	879
	unresolved.	880
	5. Justification: Provide clear and concise rea-	881
	soning that cites specific elements from the SEC	882
	comment, company response, and amended evi-	883
	dence. Highlight specific unmet requirements or	884
	gaps in the evidence that contribute to an unre-	885
	solved classification.	886
	6. Feedback Incorporation: Pay attention to feed-	887
	back from previous examples to identify common	888
	pitfalls, such as failing to compare the SEC re-	889
	quest against the evidence adequately or missing	890
	key elements in the response.	891
	B Additional Dataset Details	892
	The most common issue categories are other disclo-	893
	sure issues (183), risk factors (91), SPAC disclosure	894

895 (78), revenue recognition (75), crypto (28), non-
896 GAAP (16), and cybersecurity (1). The most com-
897 mon gap types are resolved visible evidence (153),
898 missing specific disclosure (140), missing quantifi-
899 cation (59), resolved-but-regulator-requested-more-
900 detail (38), missing exhibit or document (29), in-
901 complete accounting analysis (25), incomplete le-
902 gal or regulatory analysis (25), and other visible
903 evidence gap (3).

904 **C Additional Qualitative Cases**

905 The qualitative appendix used during analysis in-
906 cludes full SEC comments, company responses,
907 amended evidence snippets, method predictions,
908 GEPA-full rationales, and later SEC follow-up text
909 for selected cases. These include stablecoin termi-
910 nology, financial forecast assumptions, PIPE price-
911 per-share disclosure, Item 404 related-party disclo-
912 sure, expense breakout and trend discussion, and
913 SPAC transaction-background disclosure.